

CLINICAL PRESENTATION

A specific diagnosis in an individual or family with ichthyosis can help predict prognosis and is essential for genetic counseling. Several clinical features aid in distinguishing between different forms of ichthyosis, including:

- Age of onset
- Presence of a collodion membrane at birth
- Quality of scale (e.g., fine vs. coarse)
- Presence or absence of erythroderma
- Skin abnormalities in specific areas (e.g., palms, soles, ectropion, eclabium)
- Adnexal involvement (e.g., alopecia, hair shaft abnormalities)
- Extracutaneous manifestations or systemic involvement

The appearance of the skin surface varies among types of ichthyosis. Visible scaling—ranging from fine to thick, plate-like flakes—may be present. Some patients exhibit thickened skin with or without obvious scaling. The term **hyperkeratosis** is used clinically to describe thickening of the stratum corneum, which may or may not be associated with visible scale.

A family pedigree can clarify the inheritance pattern. Parental consanguinity suggests autosomal recessive inheritance, although many autosomal dominant forms (e.g., **epidermolytic hyperkeratosis**) arise from spontaneous mutations, so a negative family history does not exclude this mode of inheritance.

Light microscopy is diagnostic in **epidermolytic hyperkeratosis**, showing characteristic cytoplasmic vacuolization and clumping of keratin filaments. Histopathology may also be helpful in select conditions such as **Refsum disease**, **neutral lipid storage disease**, and **acquired ichthyosis in sarcoidosis**. However, in many ichthyoses, histology lacks specificity. In such cases, genetic testing can confirm the diagnosis.

The development of ichthyosis in adulthood should prompt evaluation for underlying systemic disease, as it may be a cutaneous marker of malignancy, infection, or autoimmune disorders.

ETIOLOGY AND PATHOGENESIS

The stratum corneum, the terminal product of epidermal differentiation, functions as a protective barrier. It consists of corneocytes ("bricks") embedded in a lipid-rich extracellular matrix ("mortar"). Corneocytes are protein-dense, while the intercellular matrix is composed of hydrophobic lipid bilayers rich in ceramides, cholesterol, and free fatty acids.

The structural integrity and hydration of the skin depend on the corneocytes, whereas the lipid matrix is primarily responsible for the permeability barrier that prevents transepidermal water loss. Normally, desquamation occurs in a controlled, invisible process where individual corneocytes detach and shed singly.

In ichthyosis, this balance is disrupted. There is abnormal scale formation—both in quality and quantity—compromised barrier function, and altered epidermal kinetics. Hyperkeratosis results from either increased production of corneocytes, delayed desquamation, or both.

Epidermal differentiation involves multiple coordinated steps, and defects at various stages can lead to similar clinical phenotypes. Known genetic defects include:

- **Mutations in *KRT1* or *KRT10*:** Cause **epidermolytic hyperkeratosis**, characterized by abnormal keratin filament assembly and cytolysis in the suprabasal layers.
- **Mutations in *TGM1*:** Encode transglutaminase 1, an enzyme critical for cross-linking the cornified envelope and attaching ceramides. Deficiency

leads to **lamellar ichthyosis**, a severe form of autosomal recessive congenital ichthyosis (ARCI).

- **Deficiency of steroid sulfatase:** Results in **X-linked recessive ichthyosis**, where impaired hydrolysis of cholesterol sulfate disrupts corneocyte desquamation, leading to scale accumulation.

These molecular insights underscore the complexity of epidermal differentiation and highlight how disruptions in structural proteins, enzymes, or lipid metabolism can converge on a common clinical phenotype—ichthyosis.